

 HACKATHON PITCH • AGENTS THAT ACT

Autonomous QA Orchestrator

Spec-to-PR predictive testing, isolated sandbox verification, and human-in-the-loop release gating powered by TrueForge.

 TrueForge Agent Harness  GitHub Actions CI  Linear Workflow

The Developer-as-QE Dilemma



Bloated CI Cycles

Running entire suites on minor PR changes.



Zero QA Bandwidth

Engineers shoulder 100% of test planning.



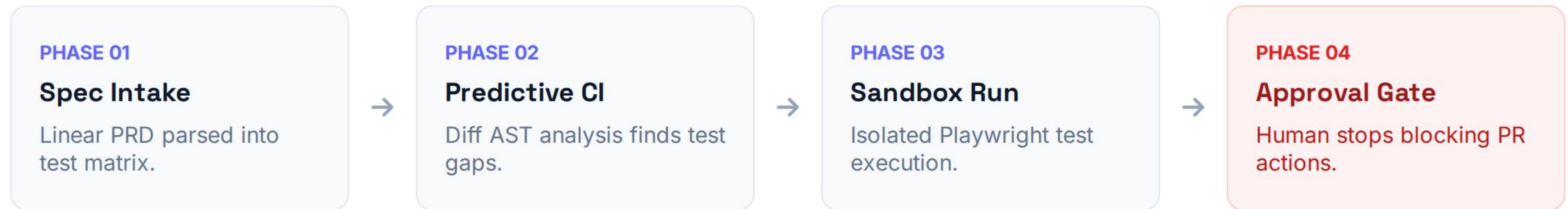
Spec Drift

Critical PRD edge cases missed in review.



Solution: An autonomous agent that reads specs, predicts impact, and synthesizes missing tests.

Four-Stage Agent Lifecycle



Legacy Test Runs vs Predictive Selection

❌ Brute-Force Pipeline

Execution: Executes full monolithic test suite.

Latency: 30+ minute turnaround on commits.

Gap Discovery: Zero coverage delta detection.

Wasteful compute on untouched features.

✅ Predictive Agent Engine

Impact Filter: Runs only touched dependency trees.

Velocity: Sub-3 minute feedback in CI.

Synthesis: Auto-generates gap-filling specs.

Pinpoints regressions before code merges.

Isolated Sandboxing & Output Artifacts

```
// CI Runner Artifact Generation  
  
$ cat run-tests.sh  
pnpm exec vitest run tests/checkout.spec.ts || true  
pnpm exec playwright test tests/pr-gap.spec.ts || true  
  
$ cat coverage-analysis.json  
{ "total_changed": 3, "uncovered": ["coupon.ts"] }
```

Zero Repo Mutation

Never edits existing production files.

Structured Confidence

AI confidence matrix scored out of 100.

Deterministic Scripts

Standard shell commands for native test runners.

Stopping Before Irreversible Actions



TrueForge Human Checkpoint

Test failure identified in sandbox. Destructive actions proposed:

1. Block PR Merge Check

2. File High-Sev Linear Bug

Zero irreversible actions executed without explicit human sign-off.

Gate Enforced

Built for "Agents That Act"



Real Systems

Integrates Linear, GitHub, and Sentry.



Safe Sandbox

Runs Playwright in containerized isolation.



Stop Checkpoint

Prompts developer before filing bugs.



TrueForge Native

Built on TrueFoundry agent harness.

 Live Demonstration

Autonomous QA in Action

From Linear requirement to CI verification and human approval gate in under 3 minutes.

github.com/team/autonomous-qa-agent